Problem Set 2

Applied Stats/Quant Methods 1

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1]Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

Step 1: assumptions

- categorical variables: class and police behaviour (whether they stopped the drivers and requested a bribe from them)
- the observations are independent
- the frequency in (most) cells is over 5

Step 2: hypotheses

- null hypothesis: the variables (class and police behaviour) are statistically independent.
- alternative hypothesis: the variables (class and police behaviour) are statistically dependent.

Step 3: calculate the test statistic

- set up the data and create a contingency table

```r
observedfs <- matrix(c(14, 7, 6, 7, 7, 1),
          nrow = 2)

# add row and column names
rownames(observedfs) <- c("Upper Class", "Lower Class")
colnames(observedfs) <- c("Not stopped", "Bribe requested",
            "Stopped/given warning")

# add margins - total counts for each category
con_table = addmargins(observedfs)
```

```
            Not stopped Bribe requested Stopped/given warning Sum
Upper Class          14               6                     7  27
Lower Class           7               7                     1  15
Sum                  21              13                     8  42
```

- extract the grand total from the contingency table
- calculate the expected frequencies using the formula :
  expected frequency = row total * column total / grand total

```r
# get the grand total from the sum column
grand_total = con_table[3,4]

# calculate the expected frequencies
# row total * column total / grand total
ef1 = con_table[1,4] * con_table[3,1]/ grand_total
ef2 = con_table[2,4] * con_table[3,1]/ grand_total
ef3 = con_table[1,4] * con_table[3,2]/ grand_total
ef4 = con_table[2,4] * con_table[3,2]/ grand_total
ef5 = con_table[1,4] * con_table[3,3]/ grand_total
ef6 = con_table[2,4] * con_table[3,3]/ grand_total
```

- create a matrix containing the expected frequencies

```r
expectedfs <- matrix(c(ef1, ef2, ef3, ef4, ef5, ef6),
                     nrow = 2)

rownames(expectedfs) <- c("Upper Class", "Lower Class")
colnames(expectedfs) <- c("Not stopped", "Bribe requested",
                   "Stopped/given warning")
expectedfs
```

```
            Not stopped Bribe requested Stopped/given warning
Upper Class        13.5        8.357143              5.142857
Lower Class         7.5        4.642857              2.857143
```

- having both the observed and the expected frequencies, we can calculate the chi-square test statistic :

```r
chi_square <- sum((observedfs - expectedfs)^2 / expectedfs)
chi_square
```

```
[1] 3.791168
```

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?
Step 4: calculate the p-value

```r
# calculate the degrees of freedom
# df = (r - 1) * (c - 1)
df = (2-1)*(3 -1)

# calculate the p-value
p = pchisq(chi_square, df, lower.tail = FALSE)
p
```

```
[1] 0.1502306
```

[2] Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

Step 5: conclusion

The p-value is higher than the alpha value (0.1). Thus, we cannot reject the null hypothesis that the variables class and police behaviour are statistically independent. The test result is not statistically significant.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.3220306	-1.641957	1.523026
Lower class	-0.3220306	1.641957	-1.523026

```
# get the standardised residuals
result <- chisq.test(observedfs)
```

(d) How might the standardized residuals help you interpret the results?

Since none of the standardized residuals have an absolute value higher than 2, we can conclude that there are no substantial differences from what we would expect. In this case, the values of the standardised residuals suggest that police officers' bevahiour did not vary significantly based on the class of the drivers they interacted with. The values of the standardised residuals also indicate that we are less likely to be able to reject the null hypothesis, which aligns with the conclusion of the hypothesis test conducted previously.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
`GP`	An identifier for the Gram Panchayat (GP)
`village`	identifier for each village
`reserved`	binary variable indicating whether the GP was reserved for women leaders or not
`female`	binary variable indicating whether the GP had a female leader or not
`irrigation`	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
`water`	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

Null hypothesis: $\beta = 0$
The reservation policy is not associated with the number of new/repaired drinking water facilities in the villages.

Alternative hypothesis: $\beta \neq 0$
The reservation policy is associated with the number of new/repaired drinking water facilities in the villages.

With β being the slope - the coefficient estimate for the reservation policy.

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
df <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/master/
   PREDICTION/women.csv", header=T)
head(df)

# fit the linear model
# outcome = water, explanatory = reserved
fit = lm(df$water ~ df$reserved)
summary(fit)
```

```
Call:
lm(formula = df$water ~ df$reserved)

Residuals:
Min      1Q  Median      3Q     Max
-23.991 -14.738  -7.865   2.262 316.009

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.738      2.286   6.446 4.22e-10 ***
df$reserved    9.252      3.948   2.344   0.0197 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.45 on 320 degrees of freedom
Multiple R-squared:  0.01688, Adjusted R-squared:  0.0138
F-statistic: 5.493 on 1 and 320 DF,  p-value: 0.0197
```

(c) Interpret the coefficient estimate for reservation policy.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.738      2.286   6.446 4.22e-10 ***
df$reserved    9.252      3.948   2.344   0.0197 *
```

Since the reserved variable is a binary one, the value of the intercept (14.738) shows the average number of new/repaired drinking-water facilities in the villages without the reservation policy. The slope or the coefficient estimate for the reservation policy indicates that, on average, villages where the policy was implemented have 9.252 more new/repaired drinking-water facilities. The p-value (0.0197) is smaller than the signficance level $\alpha = 0.05$, so we can reject the null hypothesis that the reservation policy is not associated with the number of new/repaired drinking-water facilities.